Mentor Bot

An AI-driven tutoring assistant that adapts explanations to learners' understanding levels using NLP, deep learning, and dynamic feedback loops for personalized education.

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Introduction

Smart artificial intelligence tutoring assistant.

Fits explanations to level of learners.

Uses NLP and deep learning

Learner dynamism in understanding.

Modular, extensible design

Work in progress prototype

1. Motivation & Problem Statement

- Chatbots that are not dynamic are not mobile.
- The students are different in terms of knowledge.
- Requires individual learning response.
- Majority of LLMs provide general responses.
- Absence of real-time adaptation.
- MentorBot overcomes the issue of personalization.

System Architecture Overview

- Input Feedback Analyzer Profile Context LLM Feedback.
- Six-module modular workflow
- Stored dynamically learner profile.
- Context retrieved out of knowledge base.
- Explanatory prompts of an adaptive type.
- Continuous learning feedback loop.

Models & Techniques Planned

- Base LLMs: Mistral, Phi-3, Llama-3
- NLP analyzers: Flan-T5, BART-MNLI
- Embeddings: MiniLM, e5-base-v2
- Context retrieval: FAISS / Chroma
- Streamlit for prototype UI

Data & Knowledge Sources

- Small internal knowledge base
- Domain-specific summaries and notes
- Learner profiles in JSON format
- Embedding-based content retrieval
- No dependency on any external dataset
- Data evolves with interactions

Expected Contributions

- Adaptive learning using NLP
- Learner profiling and feedback loop
- Integration of multiple open models
- Dynamic prompt generation
- Personalized learning experience
- Prototype for intelligent tutoring

Current Progress & Next Steps

- Architecture and design completed
- Folder structure implemented
- Model loading under development
- UI Designing in Streamlit
- Testing adaptive response generation
- Future: evaluation and deployment